ECE 510 CODEFEST-6

CMAN-6

Sneak paths in a Resistive Crossbar

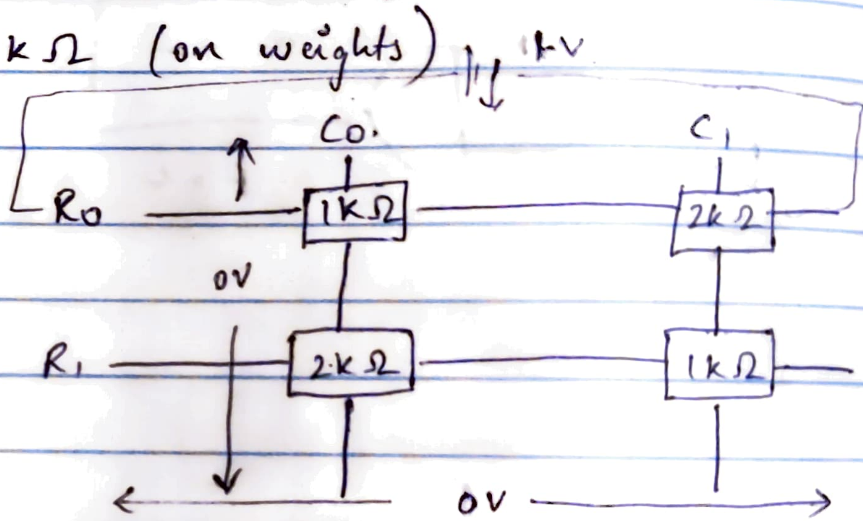
a. Given 2x2 Resistive Crossbar

$R[0][0] = 1k\Omega$ (on weights)

$R[0][1] = 2k\Omega$ (off weights)

$R[1][0] = 2k\Omega$ (off weights)

$R[1][1] = 1k\Omega$ (on weights)



$V = iR$

$$i = \frac{V}{R}$$

$$= \frac{1V}{1k\Omega}$$

$$= \frac{1}{1000} = 0.001\,A$$

$$i = 1\,mA$$

b. Given, $I_{Total} = I_{ideal} + I_{sneaks}$

$$I_{ideal} = \frac{1V - 0V}{1k\Omega} = 1\,mA$$

I_{sneaky} : Current through

$2k\Omega$, $1k\Omega$, $2k\Omega$
R_{01} · R_{11} R_{10}

Using KCL,

Current incoming = Current Outgoing

→ At $2k\Omega(R_{01})$:

$$\frac{1 - V_{col1}}{2k\Omega} = \frac{V_{col1} - V_{row1}}{1k\Omega}$$

$\Rightarrow 1 - V_{col1} = 2V_{col1} - 2V_{row1}$

$\Rightarrow 3V_{col1} = 1 + 2V_{row1}$ — 1

→ At $1k\Omega(R_{11})$:

$$\frac{V_{col1} - V_{row1}}{1k\Omega} = \frac{V_{row1} - 0}{2k\Omega}$$

$\Rightarrow 2V_{col1} - 2_{row1} = V_{row1}$

$\Rightarrow 3V_{row1} = 2V_{col1}$

$\Rightarrow V_{row1} = \frac{2}{3}V_{col1}$ — 2

$\Rightarrow 3V_{col1} = 1 + 2 \times \frac{2}{3}V_{col1}$

$\Rightarrow 3V_{col1} = 1 + \frac{4}{3}V_{col1}$

$\Rightarrow 9V_{col1} = 3 + 4V_{col1}$

$\Rightarrow 5V_{col1} = 3 \Rightarrow \boxed{V_{col1} = \frac{3}{5} = 0.6\,V}$

$\Rightarrow \boxed{V_{row1} = \frac{2}{\cancel{3}} \times \frac{\cancel{3}}{5} = \frac{2}{5} = 0.4\,V}$

c. $I_{total} = I_{ideal} + I_{sneaks}$

I_{sneaky} = Current through [$I_{ideal} = 1mA$]

$\Rightarrow I_{2k\Omega} \ (R_{01}) = \frac{1 - 0.6}{2k} = \frac{0.4}{2k}$

$= 0.2 \ mA$

$\Rightarrow I_{1k\Omega} \ (R_{11}) = \frac{0.6 - 0.4}{1k} = 0.2 \ mA$

$\Rightarrow I_{2k\Omega} \ (R_{10}) = \frac{0.4}{2k} = 0.2 \ mA$

$\Rightarrow I_{total} = I_{ideal} + I_{sneaky}$

$\Rightarrow 1mA + 0.2 \ mA$

$\Rightarrow \underline{\underline{1.2 \ mA}}$

d. The sneak path current corrupts the Matrix-Vector Multiplication (MVM) because the sensor at Col0 cannot distinguish between the ideal 1.0 mA current from the target cell and the unintended 0.2 mA leakage from neighbouring cells. This unintended "detour" caused a 20% error in the calculation by adding leakage to the total output. In large arrays, these sneak paths multiply accross thousands of devices, significantly reducing the precision of the hardware. Ultimately, this means the hardware may report incorrect values for AI computations, limiting the reliability of the entire system.